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Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A large amount of P is added to a small amount of Q. A reversible reaction occurs in which P reacts to form Q.

$$P \rightleftharpoons Q$$

Which graph shows how the concentrations of P and Q change as the reaction reaches equilibrium?

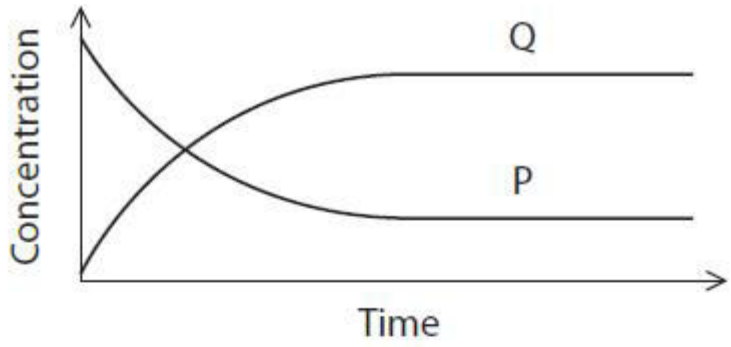
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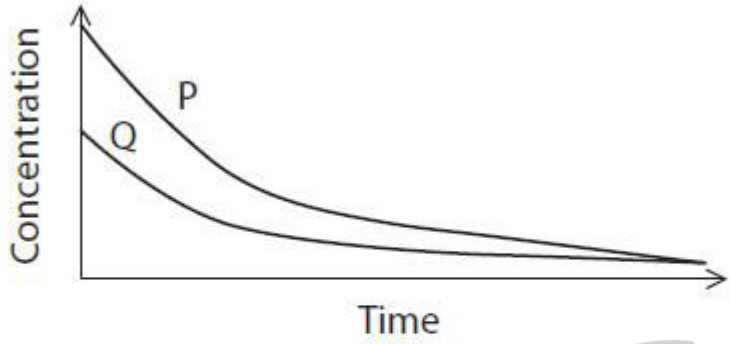
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☒ **A**

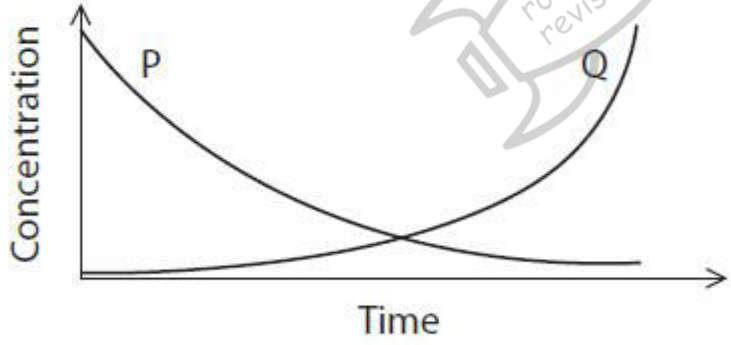


☒ **B**

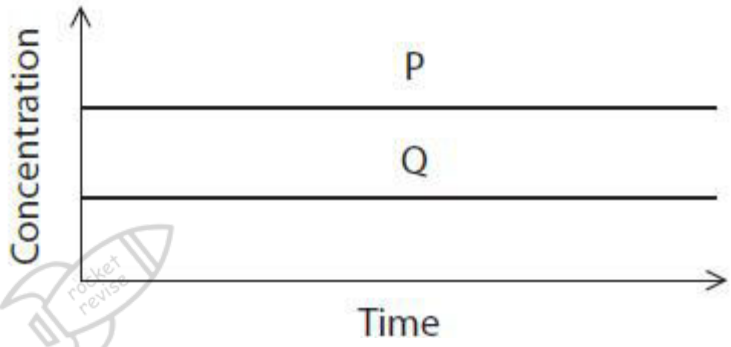


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☒ **C**



☒ **D**



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(Total for question = 1 mark)

Q2.

The equation for a reversible reaction is shown.

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$$PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g) \qquad \Delta H = -88\,kJ\,mol^{-1}$$

What effect will each change have on the rate of reaction and the equilibrium yield of phosphorus(V) chloride?

(a) Increasing the temperature at constant pressure.

(1)

	Effect on rate of reaction	Effect on yield of $PCl_5(g)$
☒ **A**	increase	decrease
☒ **B**	decrease	decrease
☒ **C**	increase	increase
☒ **D**	decrease	increase

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(b) Increasing the pressure at constant temperature.

(1)

	Effect on rate of reaction	Effect on yield of $PCl_5(g)$
☒ **A**	increase	decrease
☒ **B**	decrease	decrease
☒ **C**	increase	increase
☒ **D**	decrease	increase

(Total for question = 2 marks)

Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equilibrium shifts to the right-hand side when the pressure in the system **decreases** at constant temperature?

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☒ **A** $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

☒ **B** $F_2(g) + H_2(g) \rightleftharpoons 2HF(g)$

☒ **C** $C_6H_6(g) + 3H_2(g) \rightleftharpoons C_6H_{12}(g)$

☒ **D** $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$

(Total for question = 1 mark)

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Q4.

This question is about sulfuric acid.

(a) Sulfuric acid is manufactured in a three-step process.

Step 1 $S(s) + O_2(g) \rightarrow SO_2(g)$

Step 2 $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

Step 3 $SO_3(g) + H_2O(l) \rightarrow H_2SO_4(l)$

(i) At room temperature and pressure, sulfur is a solid and oxygen a gas.
Explain why sulfur and oxygen exist in different states by referring to the intermolecular forces involved.

(2)

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(ii) Step **2** is a reversible reaction.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad \Delta H = -196\,kJ\,mol^{-1}$$

State the effect on the position of equilibrium when the temperature and pressure are increased.
Justify your answers.

(4)

Increasing temperature at constant pressure

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Increasing pressure at constant temperature

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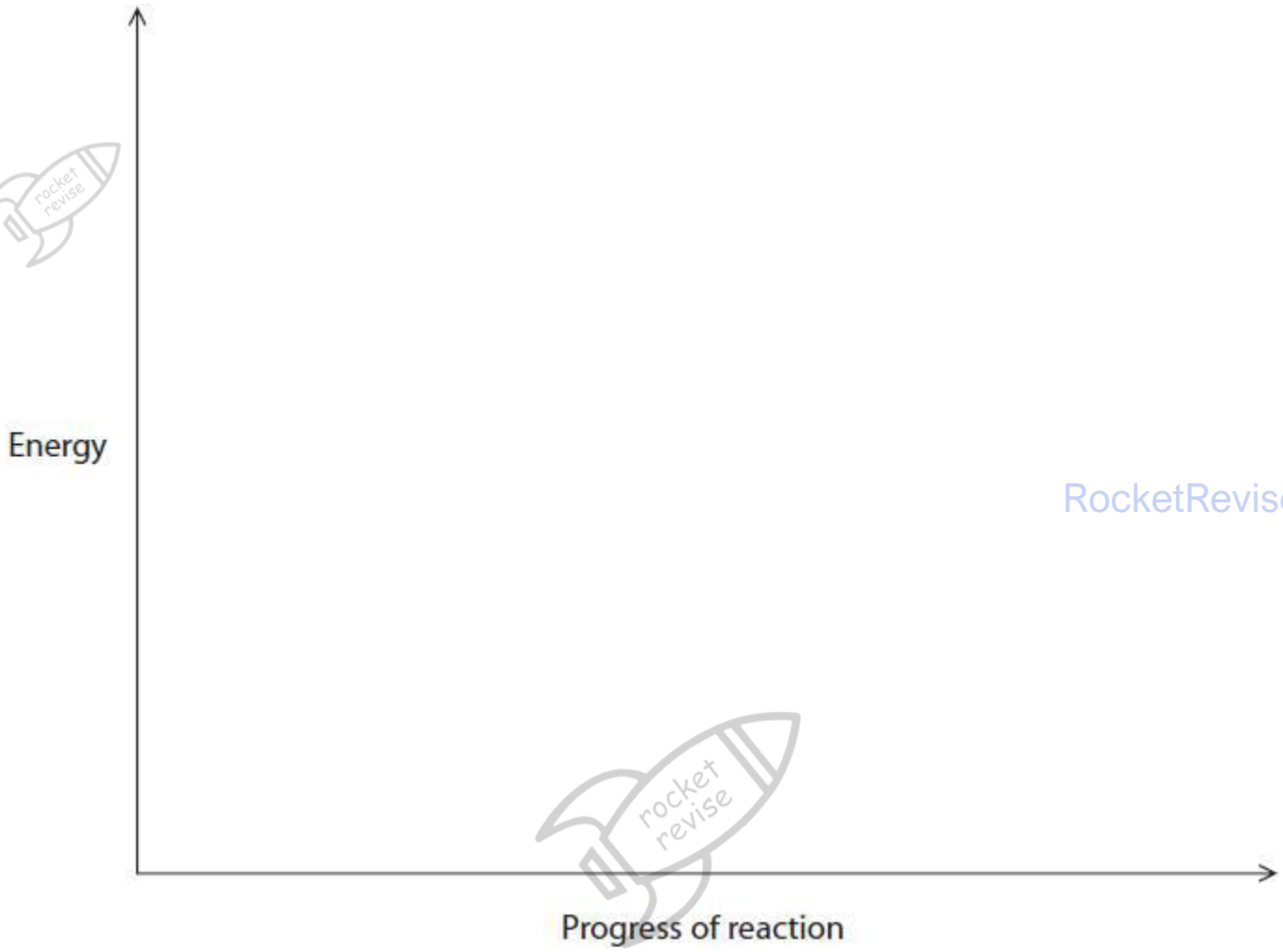
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(iii) Vanadium(V) oxide, V_2O_5, is used to catalyse the reaction in Step **2**.
Complete the reaction profile to show both the catalysed and uncatalysed reactions.
Label the activation energy and the enthalpy change. Your diagram does not have to be to scale.

(4)

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(iv) Explain how the use of a catalyst makes this reaction more sustainable.

(2)

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* (b) Concentrated sulfuric acid reacts with solid potassium chloride and with solid potassium iodide.

Compare and contrast these reactions to show the difference in reducing ability of these two halide ions.
In your answer describe what would be observed and identify the products formed.

(6)

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(c) Dilute sulfuric acid is used in many laboratories. It is often supplied as concentrated sulfuric acid and then diluted.

A technician makes 500 cm^3 of 1.5 mol dm^{-3} sulfuric acid by adding 40.5 cm^3 of concentrated sulfuric acid to distilled water.
Calculate the concentration of the **concentrated** sulfuric acid, in mol dm^{-3} .

(2)

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(Total for question = 20 marks)

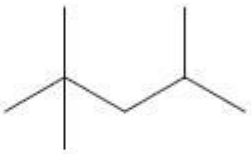
Q5.

Fuels

Fuels burn in oxygen to release a lot of energy.

Many hydrocarbons and alcohols are used as fuels. During complete combustion, they produce carbon dioxide and water.

Petrol contains 2,2,4-trimethylpentane, an isomer of octane, that promotes smooth combustion.



2,2,4-trimethylpentane

Alcohols, such as methanol and ethanol, can be used as fuels either on their own or as additives in petrol.

(a) The standard enthalpy change of combustion, $\Delta_c H^{\ominus}$, of 2,2,4-trimethylpentane is −5461 kJ mol^{-1}.

(i) State the two standard conditions for this enthalpy change.

(1)

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(ii) Write the equation for the complete combustion of 2,2,4-trimethylpentane, using **molecular** formulae.

State symbols are not required.

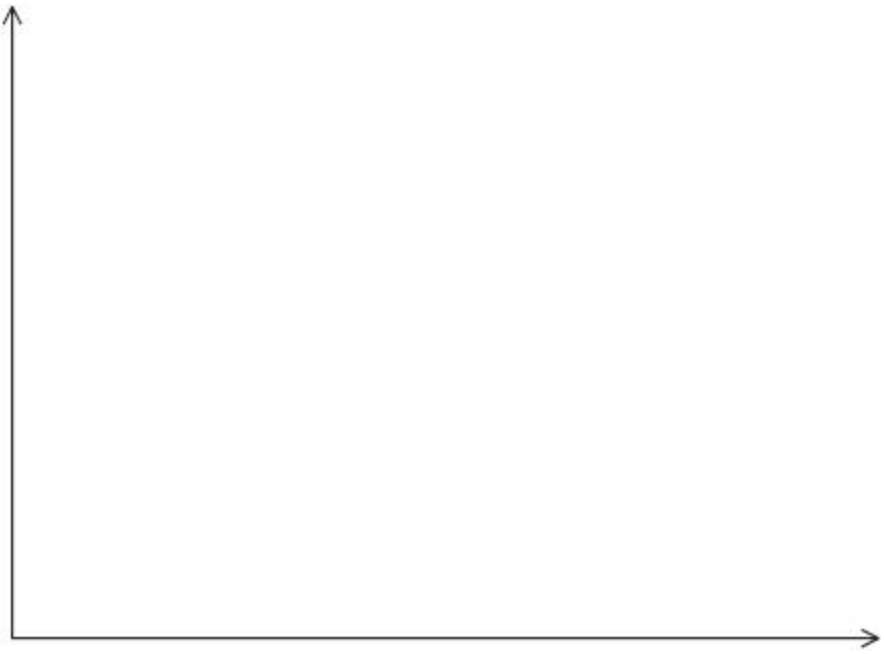
(2)

(iii) Draw a labelled enthalpy level diagram for the complete combustion of 2,2,4-trimethylpentane.

(2)

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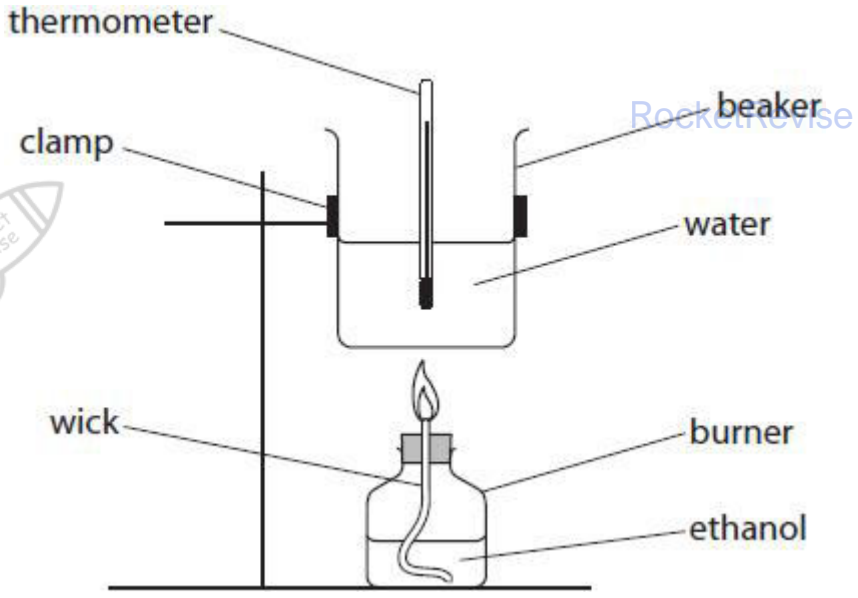


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(iv) Calculate the heat energy released during the complete combustion of 1 dm^3 of 2,2,4-trimethylpentane.
[Density of 2,2,4-trimethylpentane = 0.692 g cm^{-3}]

(3)

(b) In an experiment to determine the enthalpy change of combustion of ethanol, C_2H_5OH, a student used the apparatus shown.



Results:

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Mass of water = 100.0 g
Mass of ethanol used = 0.305 g
Temperature rise of water = 13.2°C

(i) Calculate the enthalpy change of combustion of ethanol.
Give your answer to an appropriate number of significant figures, and include a sign and units.
[Specific heat capacity of water = 4.18 J $g^{-1}°C^{-1}$]

(4)

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(ii) The uncertainty in each thermometer reading is ±0.05 °C.
Calculate the percentage uncertainty in the temperature rise in this experiment.

(1)

(iii) The student looked in a data book and found the actual value for the standard enthalpy change of combustion of ethanol was more exothermic than the experimental value obtained.
Give **two** reasons for the difference between the data book value and the experimental value, other than referring to standard conditions.

(2)

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(c) The enthalpy changes for the conversion of four compounds in the gas phase into their constituent atoms are shown.

$H_2O(g) \rightarrow 2H(g) + O(g)$ $\quad \Delta_r H = +928\ kJ\ mol^{-1}$

$CH_4(g) \rightarrow C(g) + 4H(g)$ $\quad \Delta_r H = +1740\ kJ\ mol^{-1}$

$CH_3OH(g) \rightarrow C(g) + 4H(g) + O(g)$ $\quad \Delta_r H = +2105\ kJ\ mol^{-1}$

$C_2H_5OH(g) \rightarrow 2C(g) + 6H(g) + O(g)$ $\quad \Delta_r H = +3322\ kJ\ mol^{-1}$

Calculate the bond enthalpy of the C—C bond, in kJ mol^{-1}.

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You **must** show your working.

(3)

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(d) Ethanol can be manufactured by reacting ethene with steam.

$$C_2H_4(g) + H_2O(g) \leftrightarrow C_2H_5OH(g) \qquad \Delta_r H = -45 \text{ kJ mol}^{-1}$$

This reaction is usually carried out in industry at 300 °C and 70 atm pressure using a catalyst. Explain the effect on the equilibrium position and the equilibrium yield of ethanol if the reaction is carried out at 300°C and 200 atm pressure.

(2)

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(Total for question = 20 marks)

Q6.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A mixture of hydrogen and iodine is left in a sealed container at 300°C until equilibrium is established.

$$H_2(g) + I_2(g) \leftrightarrow 2HI(g)$$

The equilibrium mixture is then cooled and the colour of the mixture darkens.

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Which is correct?

	Change in position of equilibrium when the system is cooled	Enthalpy change of forward reaction
☒ **A**	left	endothermic
☒ **B**	left	exothermic
☒ **C**	right	endothermic
☒ **D**	right	exothermic

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(Total for question = 1 mark)

Q7.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Nitrogen dioxide and dinitrogen tetroxide exist in equilibrium.

$$\underset{\text{brown gas}}{2NO_2(g)} \rightleftharpoons \underset{\text{colourless gas}}{N_2O_4(g)}$$

When an equilibrium is set up in a gas syringe, the mixture is pale brown.

When the mixture is compressed the colour becomes

☒ **A** darker
☒ **B** lighter
☒ **C** darker and then lighter
☒ **D** lighter and then darker

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(Total for question = 1 mark)

Q8.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which conditions give the highest yield for the forward reaction?

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$NH_4Cl(s) \leftrightarrow NH_3(g) + HCl(g)$ $\Delta H = +176$ kJ mol^{-1}

- **A** high temperature, high pressure
- **B** high temperature, low pressure
- **C** low temperature, high pressure
- **D** low temperature, low pressure

(Total for question = 1 mark)

Q9.

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This question is about carbon dioxide.

(a) (i) Carbonate ions may be identified by their reaction with aqueous acid to produce carbon dioxide. Give the **ionic** equation for this reaction. Include state symbols.

(2)

(ii) Limewater (an aqueous solution of calcium hydroxide) can be used to test for the presence of gaseous carbon dioxide.
State what would be **seen** when carbon dioxide is bubbled through limewater.

(1)

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(b) A student determined the solubility of calcium hydroxide in water by titration with hydrochloric acid. The equation for the reaction is shown.

$$Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$$

Procedure

- calcium hydroxide powder was added to distilled water and the mixture was stirred until no more solid dissolved.
The excess solid was filtered off to leave a saturated solution.

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- 25.0 cm^3 portions of this solution were titrated with 0.0500 mol dm^{-3} hydrochloric acid with phenolphthalein as an indicator until two concordant titres were obtained.

The mean titre was 18.95 cm^3.

(i) Calculate the concentration of the saturated solution, in g dm^{-3}.

(4)

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(ii) The student used the same procedure to determine the solubility of strontium hydroxide.
Explain whether or not the mean titre value for strontium hydroxide would be different from that for calcium hydroxide.

(2)

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(c) Between 1960 and 2020 the amount of carbon dioxide in the atmosphere rose from 320 parts per million (ppm) to 420 ppm.

The recent rapid increase in the atmospheric carbon dioxide is affecting the chemistry of seawater.
Carbon dioxide dissolves in water to form carbonic acid, $H_2CO_3(aq)$.
The carbonic acid dissociates in water according to the equilibrium shown.

$$H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$$

Explain, in terms of the equilibrium, the effect of the increase in atmospheric carbon dioxide on the acidity of seawater.

(3)

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(Total for question = 12 marks)

Q10.

The equilibrium reaction shown in the equation was studied by placing the components into a sealed glass container.

$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

At equilibrium, which of the following statements is **not** true?

(1)

☒ **A** the concentrations of the $NO_2(g)$ and $N_2O_4(g)$ both remain constant

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- ☒ **B** the total number of molecules is constant
- ☒ **C** the forward and reverse reactions have both stopped
- ☒ **D** the rate of the forward reaction is equal to the rate of the reverse reaction

(Total for question = 1 mark)

Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which statement is **not** true for a system at dynamic equilibrium?

- ☒ **A** the position of equilibrium is affected by temperature, by pressure and by catalysts
- ☒ **B** the concentration of reactants is constant
- ☒ **C** the forward and backward reactions occur at equal rates
- ☒ **D** the equilibrium can be reached from either direction

(Total for question = 1 mark)

Q12.

This question is about Group 7 hydrides.

(a) The table shows the boiling temperatures of some Group 7 hydrides.

Hydrogen halide	Boiling temperature / °C
HCl	−85
HBr	−67
HI	−35

The increase in boiling temperature in these hydrogen halides is due to London forces.

(i) Describe how London forces arise.

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(ii) State why the London forces in hydrogen iodide are stronger than those in hydrogen chloride and hydrogen bromide.

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(iii) The boiling temperature of hydrogen fluoride is +20 °C.
State what causes hydrogen fluoride to have such an unusually high boiling temperature.

(1)

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(b) Chlorine reacts with water as shown.

$$Cl_2(aq) + H_2O(l) \rightleftharpoons HOCl(aq) + HCl(aq)$$

(i) Explain why this is a disproportionation reaction by referring to relevant oxidation numbers.

(3)

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(ii) State and explain what happens to the position of equilibrium when sodium hydroxide solution, NaOH(aq), is added to this equilibrium mixture.

(2)

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(Total for question = 9 marks)

Q13.

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One industrial preparation of ethanol is by the catalysed hydration of ethene via a reversible reaction.

$$H_2O + CH_2{=}CH_2 \rightleftharpoons CH_3CH_2OH \qquad \Delta_r H = -42\ kJ\ mol^{-1}$$

(a) The hydration of ethene requires a catalyst which is coated onto a solid silicon dioxide support.

Identify, by name or formula, the catalyst used.

(1)

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(b) Under these conditions, the conversion of ethene to ethanol is 5 % at equilibrium but the overall yield of ethanol for the process is 95 %.

Suggest what happens to the equilibrium mixture containing ethanol so that the overall yield of ethanol becomes 95 %.

(2)

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(c) The temperature used is 300 °C and the pressure used is 7 MPa. These, and the amount of water present in the reaction mixture, are all compromises.

(i) Explain why the temperature used is a compromise.

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(ii) Explain why a manufacturer might suggest a pressure of greater than the compromise pressure of 7 MPa.

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(iii) At pressures greater than 7 MPa, a waxy solid forms which melts over a range of temperatures.

Identify, by name or formula, this solid.

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(1)

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(iv) Give **one** other reason for not operating the process at a pressure greater than 7 MPa. Do not include safety considerations in your answer.

(1)

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(v) Increasing the amount of water in the reaction mixture, without changing the pressure, might also be suggested.

Suggest **one** advantage and **one** disadvantage of increasing the amount of water.

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(2)

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(Total for question = 11 marks)

Q14.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The equilibrium between methane and steam is shown.

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$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g) \qquad \Delta H = +206\,kJ\,mol^{-1}$$

(a) What would be the effect of **lowering** the temperature?

- ☐ **A** rate increases, yield increases
- ☐ **B** rate increases, yield decreases
- ☐ **C** rate decreases, yield increases
- ☐ **D** rate decreases, yield decreases

(b) What would be the effect of **increasing** the pressure?

- ☐ **A** rate increases, yield increases
- ☐ **B** rate increases, yield decreases
- ☐ **C** rate decreases, yield increases

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☒ **D** rate decreases, yield decreases

(c) Some bond enthalpies are shown.

Bond	Bond enthalpy / kJ mol^{-1}
H—H	436
C≡O	1077
O—H	464

What is the bond enthalpy, in kJ mol^{-1}, of the C―H bond?

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(1)

☒ **A** 198
☒ **B** 313
☒ **C** 416
☒ **D** 532

(Total for question = 3 marks)

Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

In the production of sulfuric acid, sulfur(IV) oxide reacts with oxygen to form sulfur(VI) oxide.

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

In a closed system, the mixture is allowed to reach equilibrium.

(a) Which of the following statements must be correct?

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(1)

☒ **A** the reactions stop
☒ **B** the sum of the masses of sulfur(IV) oxide and oxygen is equal to the mass of sulfur(VI) oxide
☒ **C** the rate of formation of sulfur(VI) oxide is equal to the rate of its decomposition
☒ **D** the concentration of sulfur(IV) oxide is equal to the concentration of sulfur(VI) oxide

(b) How does the use of a catalyst affect the equilibrium?

(1)

☒ **A** only the rate of the forward reaction increases
☒ **B** the position of equilibrium moves to the right
☒ **C** the position of equilibrium moves to the left
☒ **D** there is no effect on the position of equilibrium

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(Total for question = 2 marks)

Q16.

Magnesium carbonate has been studied recently as a potential material for the removal of carbon dioxide from the atmosphere.

(a) The value of n in a sample of a hydrated magnesium carbonate, $MgCO_3 \cdot nH_2O$, can be determined by the reaction with hydrochloric acid.

The equation for the reaction is shown.

$$MgCO_3 \cdot nH_2O(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + CO_2(g) + (n+1)H_2O(l)$$

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Procedure

Step 1 Add 0.0600 mol of hydrochloric acid (an excess) to a 2.35 g sample of the hydrated magnesium carbonate.

Step 2 When the reaction is complete, transfer the resulting solution to a 250 cm^3 volumetric flask, make up to the mark with deionised water, and mix thoroughly.

Step 3 Titrate 25.0 cm^3 portions of the solution with sodium hydroxide solution of concentration 0.0960 mol dm^{-3}.

Data

The mean titre volume = 27.15 cm^3

Determine the value of n in the hydrated salt.

(7)

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(b) The enthalpy change for the thermal decomposition of anhydrous magnesium carbonate, $\Delta_r H$, was measured indirectly using the enthalpy changes for the two reactions shown and applying Hess's Law.

Reaction **1**

$MgO(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l)$ $\Delta_r H_1$

Reaction **2**

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$MgCO_3(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l) + CO_2(g)$ $\Delta_r H_2 = +3.58$ kJ mol^{-1}

An experiment was carried out to measure the enthalpy change of reaction, $\Delta_r H_1$, for Reaction **1**.

1.92 g of magnesium oxide was added, with stirring, to 40.0 cm^3 dilute hydrochloric acid (an excess) in an insulated container.

The temperature rise was 30.8 °C.

(i) Calculate $\Delta_r H_1$ for Reaction **1**. Include a sign and units in your answer.

[Assume the specific heat capacity of the solution = 4.18 J g^{-1} °C^{-1}
the density of the solution = 1.00 g cm^{-3}]

(3)

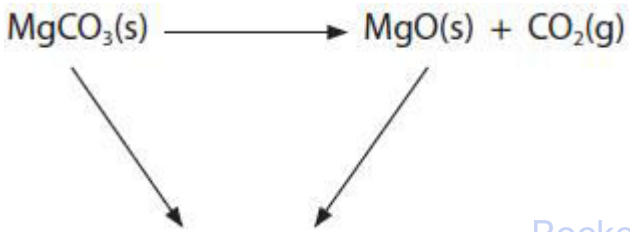
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(ii) The enthalpy change of decomposition, $\Delta_r H$, of magnesium carbonate can be found using a Hess cycle.

Calculate the enthalpy change of decomposition by using a completed Hess cycle, your answer to (b)(i) and the enthalpy change of reaction for Reaction **2**.

Give your answer to an appropriate number of significant figures.

(3)



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(iii) Give a reason why it is not possible to measure the enthalpy change for the decomposition directly.

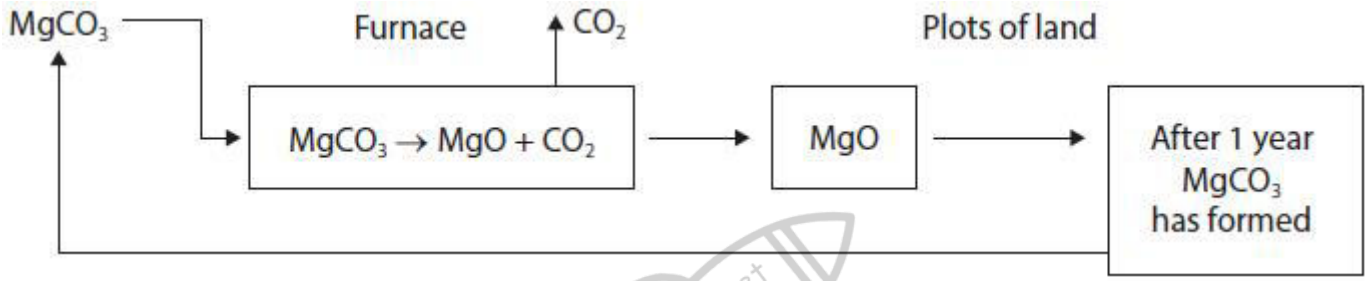
(1)

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(c) The steps involved in using magnesium carbonate ore in a process to capture carbon dioxide from the air are shown.

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- the magnesium carbonate ore is heated in a furnace and decomposes and the carbon dioxide produced is stored
- the magnesium oxide produced is transported to plots of land where it is spread to a depth of 0.1 m and reacts with carbon dioxide over a year
- the magnesium carbonate formed is then returned to the furnace and the process repeated.

Heating the furnace to decompose the magnesium carbonate accounts for most of the energy requirements of the process.

(i) Explain, with reference to the thermal stability of the Group 2 carbonates, why the use of calcium carbonate rather than magnesium carbonate would increase the cost of the process.

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(ii) Methanol can be produced from the captured carbon dioxide by reaction with hydrogen using a copper catalyst.

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The equation for the reaction is shown.

$$CO_2(g) + 3H_2(g) \rightleftharpoons CH_3OH(g) + H_2O(g) \quad \Delta H = -49.5 \text{ kJ mol}^{-1}$$

Explain why the conditions for the industrial production of methanol are a pressure of 50 atmospheres and a temperature of 250°C.

(4)

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(Total for question = 21 marks)

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